Epidemiological inferences from serological responses to cross-reacting pathogens: Simulation Study

Contents

1 Methods

To assess the performance of the model and understand its applications we conduct simulation testing across a range of scenarios. Parameter values for simulation were randomly drawn using Latin Hypercube sampling to ensure the approximately even distribution of parameter combinations across simulations. Parameter values for simulation were sampled from uniform distributions, with the ranges of values considered shown in Table S1. The ranges of possible parameter values were selected to approximately reflect antibody titer concentrations on a log relative fluorescence intensity (RFI) scale, where median fluorescence intensity (MFI) values are divided by individual-level background control values before taking the natural logarithm. The model was fit using uninformative priors on the infection prevalence and cross-reactivity parameters, while weakly informative priors were placed on the Gaussian mean and standard deviation parameters (Table S1).

Parameter	Parameter description	Simulation Limits	Prior
π	Infection prevalence	[0, 1]	Beta(1, 1)
μ_0	Mean of negative titers	[-0.5, 0.5]	Normal(0, 0.2)
μ_1	Mean increase in positive titers	[1, 3.5]	Normal(2.5, 0.5)

σ_0	SD of negative titers	[0.25, 0.5]	Normal(0.4, 0.1)
σ_1	SD of positive titers	[0.5, 0.75]	Normal(0.6, 0.1)
ϕ	Cross-reactivity	[0, 1]	Beta(1, 1)
ρ	Correlation coefficients	[0, 1]	Beta(1, 1)

Table S1 **Parameter ranges and priors for simulation study.** Ranges of possible parameter values considered for simulation using Latin Hypercube sampling. Parameters relating to antibody titers are considered on a log scale of the relative fluorescent intensity (RFI).

We first assess the performance of the model on a 2 pathogen system, simulating antibody titers against 2 related pathogens and assessing the models ability to reconstruct parameter values from 50 simulated datasets. We further assess model performance with an increasing number of pathogens, considering systems of 3, 5 and 7 related pathogens. For all simulations with ≤ 3 pathogens, we assume the Gaussian standard deviation parameters, σ_0 and σ_1 to be independent for each pathogen. For simulations with > 3 pathogens, however, we assume constant σ_0 and σ_1 across pathogens which may be a reasonable assumption when antibodies against each antigen are measured using the same assay. In each case we generate 50 simulated datasets of 1,500 individuals and fit the model to each of the simulations. All models were fit with 2 chains for 3,000 iterations in addition to 3,000 warm-up samples. Model convergence was assessed by visual inspection of chain mixing and by R-hat convergence diagnostic across all parameters. When comparing model performance we assess differences in deviance information criterion (DIC), widely applicable information criteria (WAIC) and leave-one-out expected log pointwise predictive density (ELPD LOO) metrics. When comparing the performance of models with varying numbers of Gaussian components we additionally use LIPpc and LIPpp metrics. Similar to the LIPpc metric, the likelihood increment percentage per parameter (LIPpp) penalizes the likelihood increment by the additional number of parameters used in the more complex model.

1.1 Heterogeneity in infection prevalence

In this section we consider a spatially-stratified 2 pathogen system with sampling conducted in 3 locations with varying levels of pathogen-specific infection prevalence across locations. We assume a total sample size of 1,500 individuals across the 3 locations, with 500 individuals from each. As before, Latin Hypercube sampling was used to randomly draw parameter values for 50 simulations with the same possible parameter ranges shown in Table S1. Only infection prevalence parameters were varied by location and all other parameters were assumed to be constant across locations. To each simulation we fit two model versions; a base model version where no location information is used and a single value of pathogen-specific infection prevalence is estimated for the entire population, and a second location-specific version where pathogen-specific infection prevalence is allowed to vary across the three locations.

1.2 1D mixtures

To understand the biases that can occur when antibody cross-reactivity is present but not accounted for, we fit classic single-dimension mixture models independently to antibody titer data for each pathogen using the 2 pathogen simulations. Here, each model fits a two-component Gaussian mixture distribution, with a negative and positive component as described above in the main text.

1.3 Absent pathogens

To understand the ability of the model to differentiate between pathogen presence and absence in systems of related cross-reacting pathogens, we conduct further simulations. Using the same 2 and 3 pathogen simulations from the previous section, we set the infection prevalence of one pathogen to zero and re-simulate the antibody titer data. We then fit the same base model to these new simulations as well as an "absent model" that assumes the absence of the truly absent pathogen.

2 Results

2.1 Model performance for two pathogen systems

We found high consistency between true and median estimates of infection prevalence with an adjusted R^2 of 0.99, shown in Figure S1. True values of prevalence, π, were within model 95%CrI estimates in 98% of simulations (49/50) and in 99% of prevalence estimates (99/100) (Figure S1). For the cross-reactivity parameters, ϕ, high consistency was observed between true and estimated parameter values with an adjusted R^2 of 0.83 (Figure S1). 94% of estimates (94/100) contained the true values of ϕ within the 95%CrIs, while 90% (45/50) of simulations accurately estimated both ϕ parameters. Only one simulation produced non-accurate estimates of ϕ for both parameters, the same simulation that did not accurately estimate the prevalence of pathogen A (Simulation 40). In this simulation, individuals infected with only pathogen A had higher titers against pathogen B than individuals truly infected with pathogen B, causing unidentifiability of the Gaussian components. We also note that simulations with inaccurate estimates of ϕ and/or ρ tended to occur for simulations with prevalence values close to 0% or 100%. In these simulations, 1 or more Gaussian component could not be characterised due to a limited number of data points in that component.

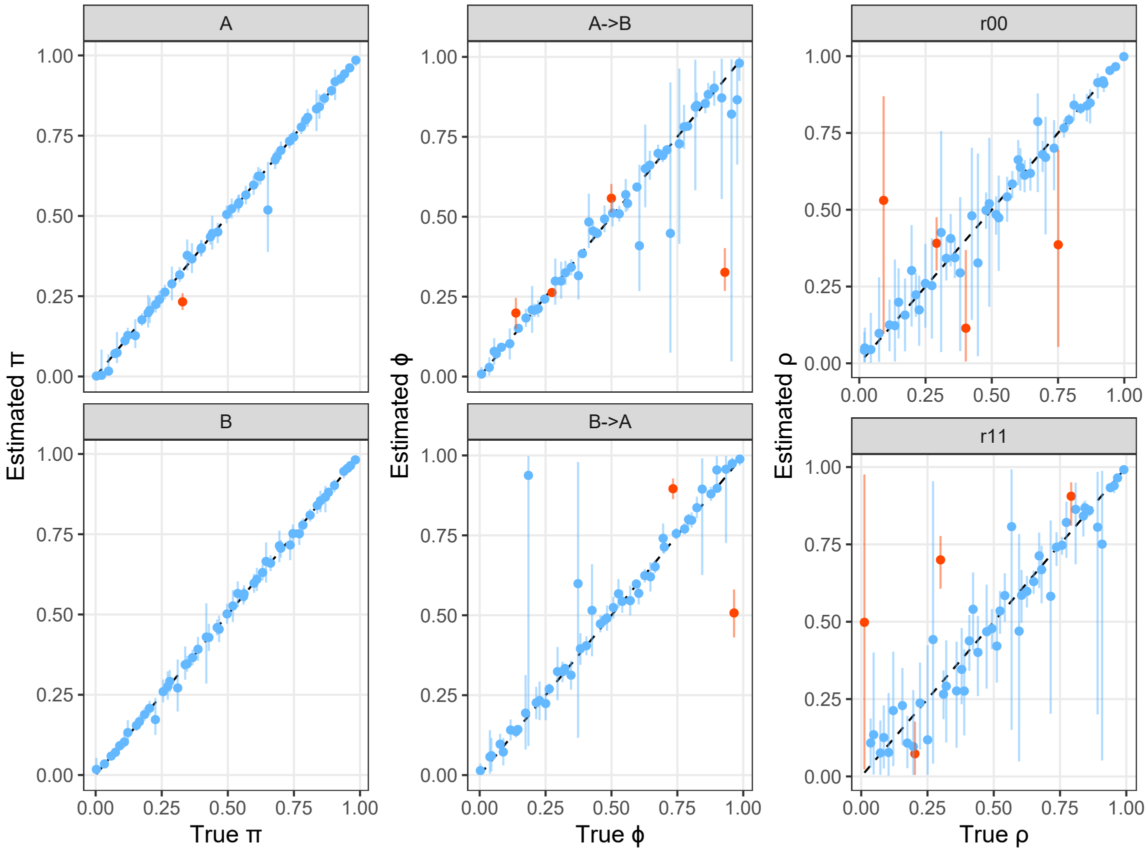


Figure S1 **Estimated vs true infection prevalence and cross-reactivity parameters in 2D simulations.** Coloured points and lines show the median and 95%CrI model estimates of infection prevalence, cross-reactivity and correlation parameters, compared to the true values in 50 simulations. The black dashed line indicates where estimated values would equal the true values. Blue points and lines highlight the model estimates where the true value was contained within the 95%CrI and orange highlight those that did not contain the true value within estimated 95%CrI. Panels A->B and B->A represent the estimates of relative titer increase from A to B and B to A, respectively. Panels r00 and r11 represent the correlation in the Gaussian distribution for individuals negative to both pathogens, ρ_{00} and positive to both pathogens, ρ_{11}, respectively.

A high level of consistency between true and estimated correlation parameters, ρ_{00} and ρ_{11}, was also observed with adjusted R^2 values of 0.88 and 0.84, respectively (Figure S1). True parameter values were within the estimated 95%CrIs for 92% and 92% of estimates for ρ_{00} and ρ_{11}. The means and standard deviations of the Gaussian components were also well identified by the model, with adjusted R^2 values of 0.96 and 0.97 between true and estimated values for μ_0 and μ_1, and of 0.84 and 0.78 for σ_0 and σ_1 parameters respectively, shown in Figure S2. True parameter values were contained within the 95%CrIs for 96% and 87% of estimates for μ_0 and μ_1 parameters and in 94% and 96% of estimates for σ_0 and σ_1 parameters (Figure S2). Model fits to the the simulated data are shown in Figure S3. We note that the model demonstrates high levels of performance even when large overlap of negative and positive titers result in unimodal titer distributions (Figure S3).

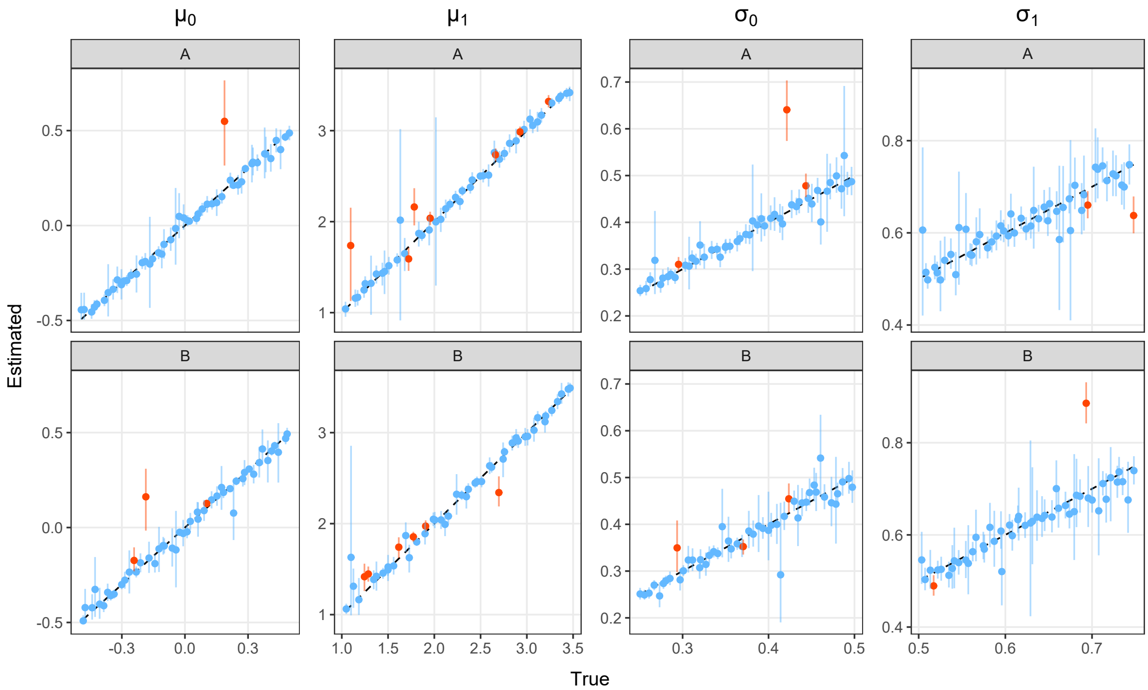
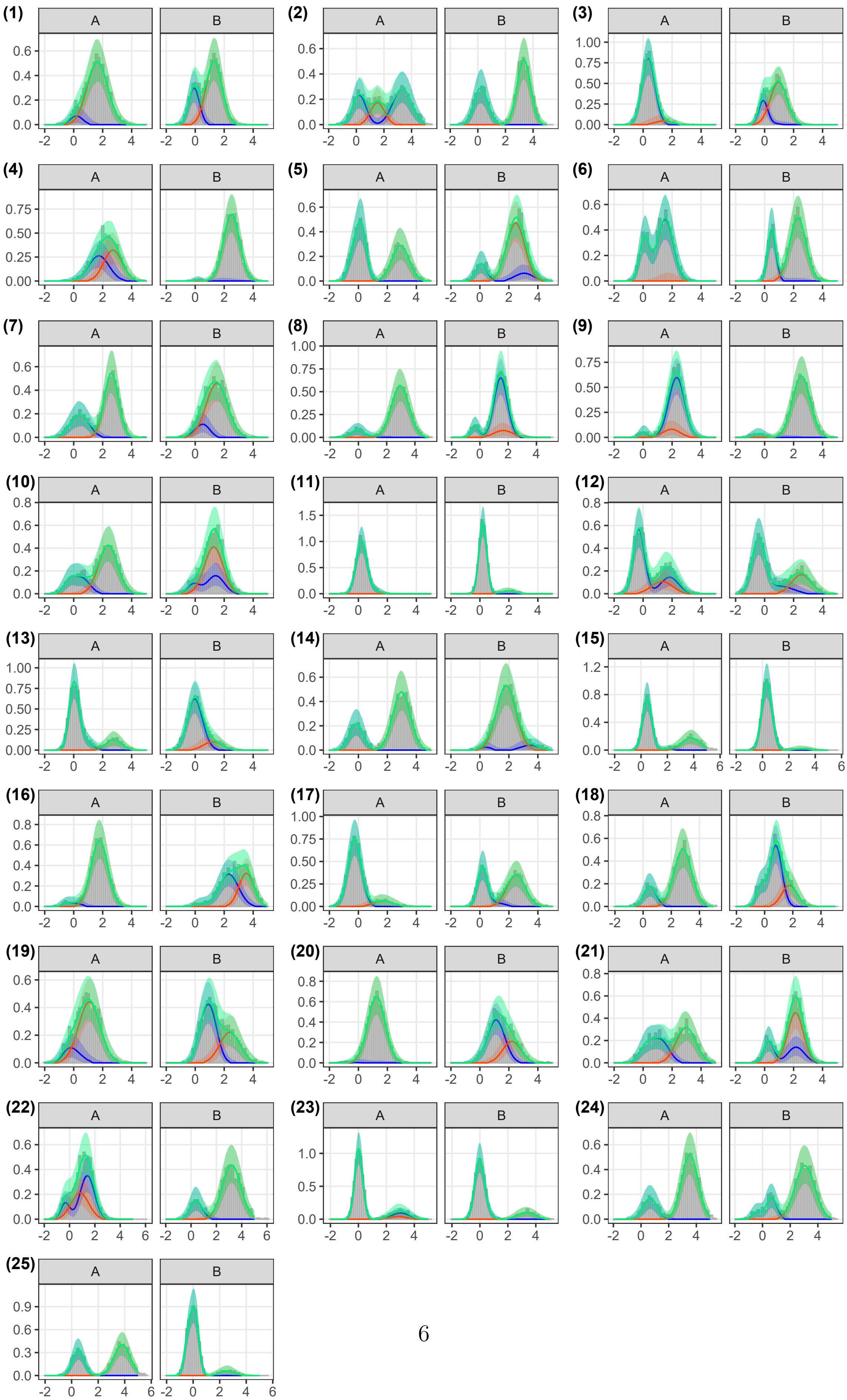


Figure S2 **Estimated vs true Gaussian mean and standard deviation parameters in 2D simulations.** Coloured points and lines show the median and 95%CrI model estimates of the μ and σ parameters, compared to the true values in 50 simulations. Each column of panels corresponds to a different parameter and each row represents estimates for pathogen A or pathogen B. The black dashed line indicates where estimated values would equal the true values. Blue points and lines highlight the model estimates where the true value was contained within the 95%CrI and orange highlight those that did not contain the true value within estimated 95%CrI.



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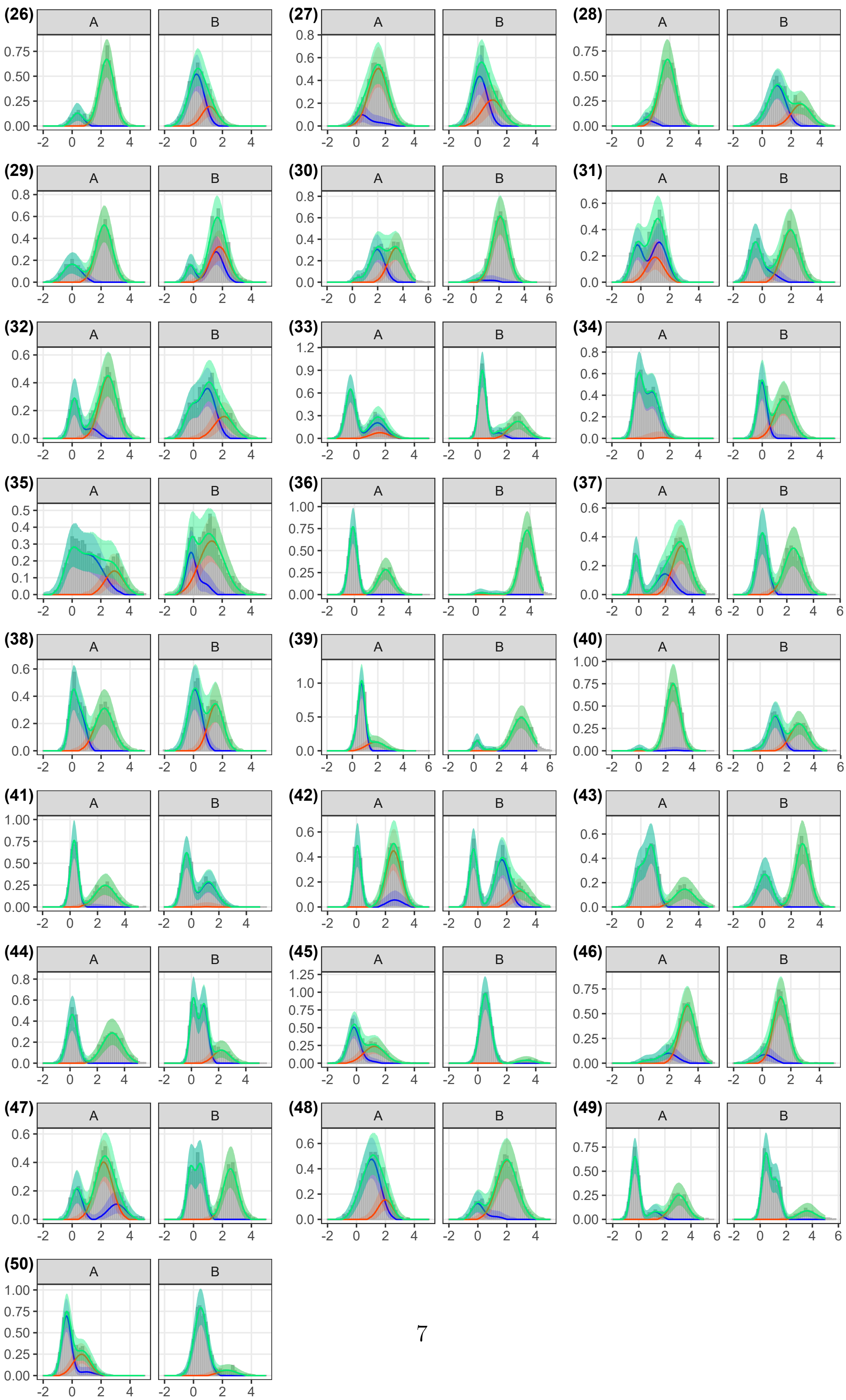
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Figure S3 **Model fits to 2D simulations.** Grey bars show the density distribution of simulated antibody titers for each of the 50 simulations, for pathogens A and B. The lines and shaded ribbons show the median and 95%CrI model reconstructed titer distributions. Blue and orange indicates the reconstructed distribution of negative and positive titers to each pathogen respectively, while green shows the overall reconstructed titer distribution (combined across all components).

2.2 Model performance with increasing pathogen numbers

We fit the model to simulated data with an increasing number of pathogens, considering systems of 3, 5 and 7 related pathogens. For each of these considered systems we simulated 50 antibody titer datasets and fit the model to each. Due to non-convergence of some models, 0, 1 and 8 model results were removed from the following summary results from the 3D, 5D and 7D simulations respectively. High consistency was observed between true and estimated prevalence with adjusted R^2 values of 0.95, 1.00 and 1.00 for 3D, 5D and 7D simulations as shown in Figure S4, panel (a). True infection prevalence values were contained within 95%CrIs in 93% of estimates (140/150 estimates) for 3D simulations, 98% of estimates (240/245) for 5D simulations and 99% of estimates (293/294) for 7D simulations (Figure S4 panel (a)). It is worth noting that the 3D simulations assumed independent σ_0 and σ_1 parameters per pathogen in line with the potential use of different assays for each pathogen. In contrast, for 5D and 7D simulations and models we assumed constant σ_0 and constant σ_1 across pathogens.

For cross-reactivity parameters, ϕ, the model showed good performance with adjusted R^2 values between estimated and true parameter values of 0.94, 0.96 and 0.98 for 3D, 5D and 7D simulations respectively, shown in Figure S4 panel (b). True values were contained within estimated 95%CrIs for 90% of estimates (270/300) in 3D simulations, 95% of estimates (928/980) in 5D simulations and 92% of estimates (1631/1764) in 7D simulations (Figure S4 panel (b) and Figure S5). Adjusted R^2 values between estimated and true values of ρ_{00} parameters were 0.95, 0.99 and 1.00 for 3D, 5D and 7D simulations respectively, with coverage of true values contained within 95%CrIs in 88% (44/50), 92% (45/49) and 93% (39/42) of simulations, shown in Figure S6 and Figure S5. For ρ_{11} parameter values, adjusted R^2 values between estimated and true were 0.88, 1.00 and 1.00 for 3D, 5D and 7D simulations with coverage of true values for 86% (43/50), 94% (46/49) and 95% (40/42) of simulations, shown in Figure S6 and Figure S5.

Estimates of μ_0 contained true parameter values within CrIs for 93% (139/150), 95% (233/245) and 91% (268/294) estimates for 3D, 5D and 7D simulations, respectively, shown in Figure S5 and Figure S6. For μ_1 parameters, estimates contained true values in 90% (135/150), 97% (238/245) and 90% (266/294) of estimates for 3D, 5D and 7D simulations (Figure S5). Estimates of σ_0 contained true parameter values within CrIs for 91% (136/150), 79% (194/245) and 100% (294/294) estimates for 3D, 5D and 7D simulations. While for σ_1, estimates contained true parameter values within CrIs for 88% (132/150), 87% (214/245) and 98% (287/294) estimates for 3D, 5D and 7D simulations. Model fits to the 3 pathogen simulations are shown in Figure S7.

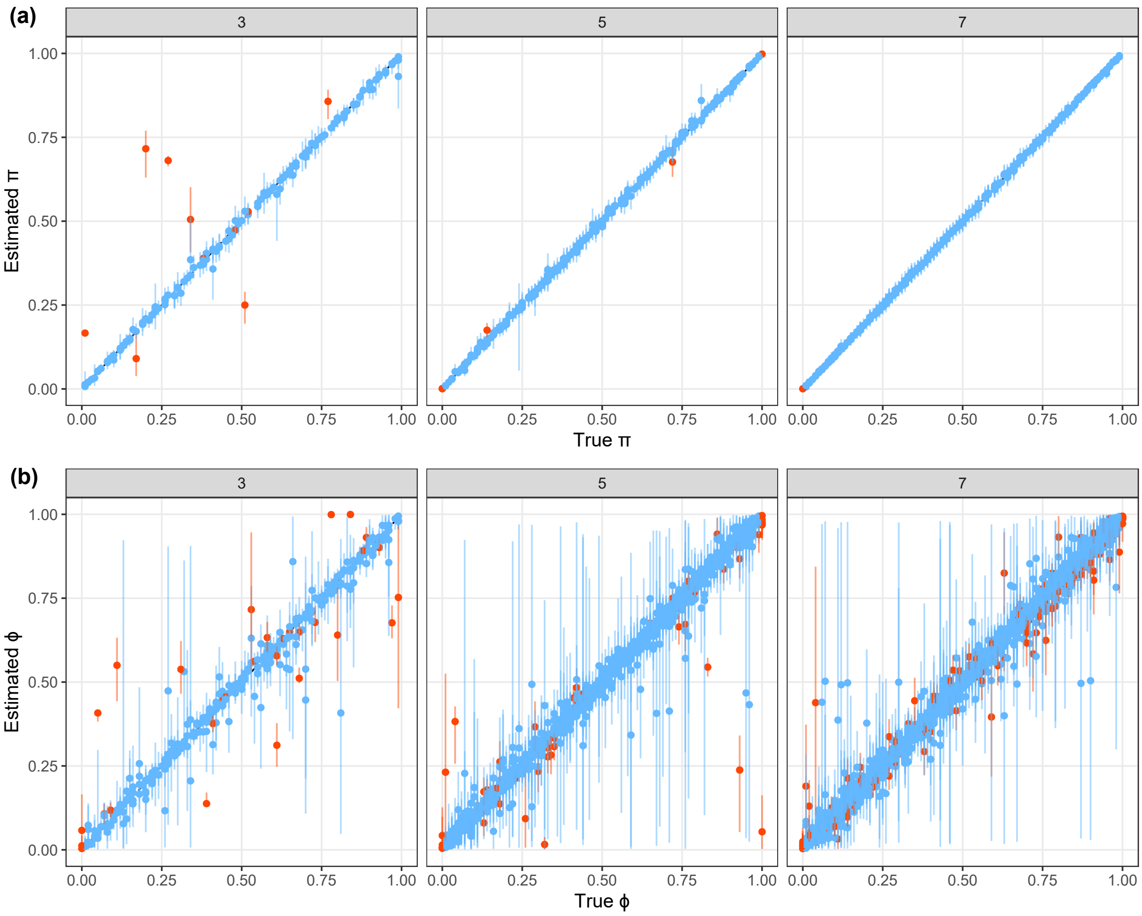


Figure S4 **Estimated vs true infection prevalence and cross-reactivity parameters in 3D, 5D and 7D simulations.** Coloured points and lines show the median and 95%CrI model estimates of (a) prevalence, π, and (b) cross-reactivity, ϕ, parameters compared to the true values in 50 simulations. Panels indicate the number of pathogens (3, 5 or 7). The black dashed line indicates where estimated values would equal the true values. Blue points and lines highlight the model estimates where the true value was contained within the 95%CrI and orange highlight those that did not contain the true value within estimated 95%CrI.

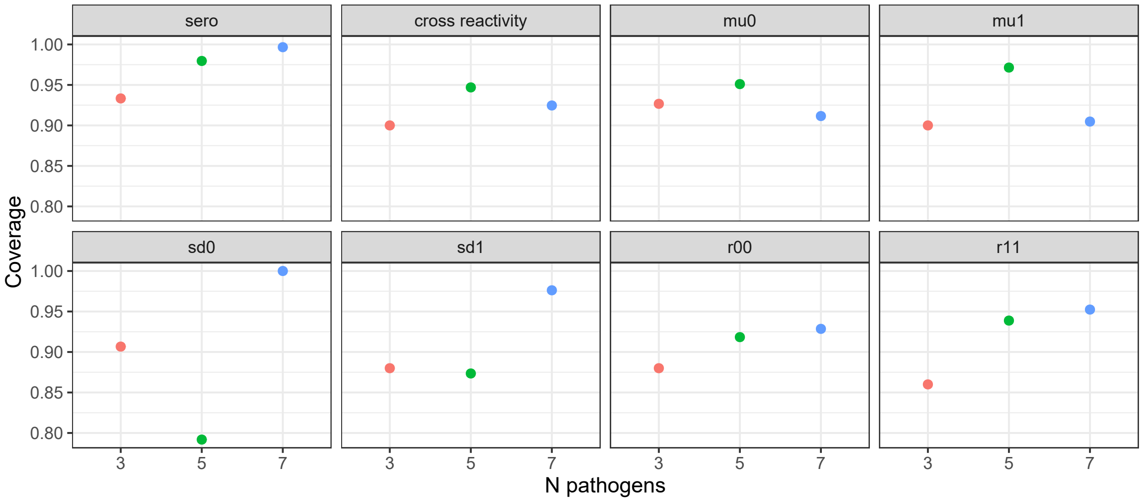


Figure S5 **Coverage of true parameter values within credible intervals in 3D, 5D and 7D simulations.** Coloured points show the proportion of estimated parameter values that contained the true value within the 95%CrI estimates for 3D, 5D and 7D simulations. Each panel represents a different parameter - infection prevalence (π), cross reactivity (ϕ), Gaussian means (μ_0 and μ_1), Gaussian standard deviations (σ_0 and σ_1), and correlation parameters (ρ_{00} and ρ_{11}).

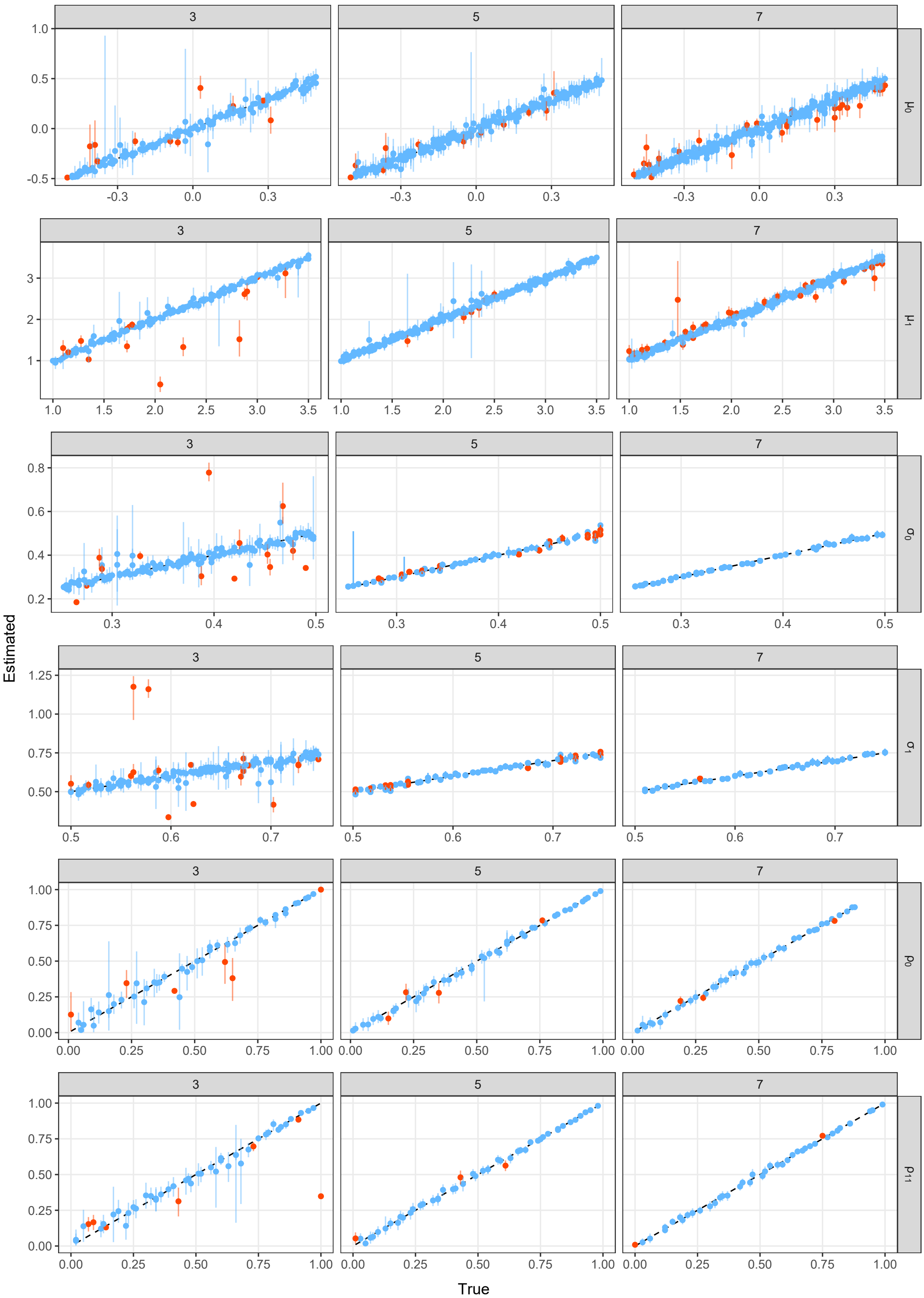
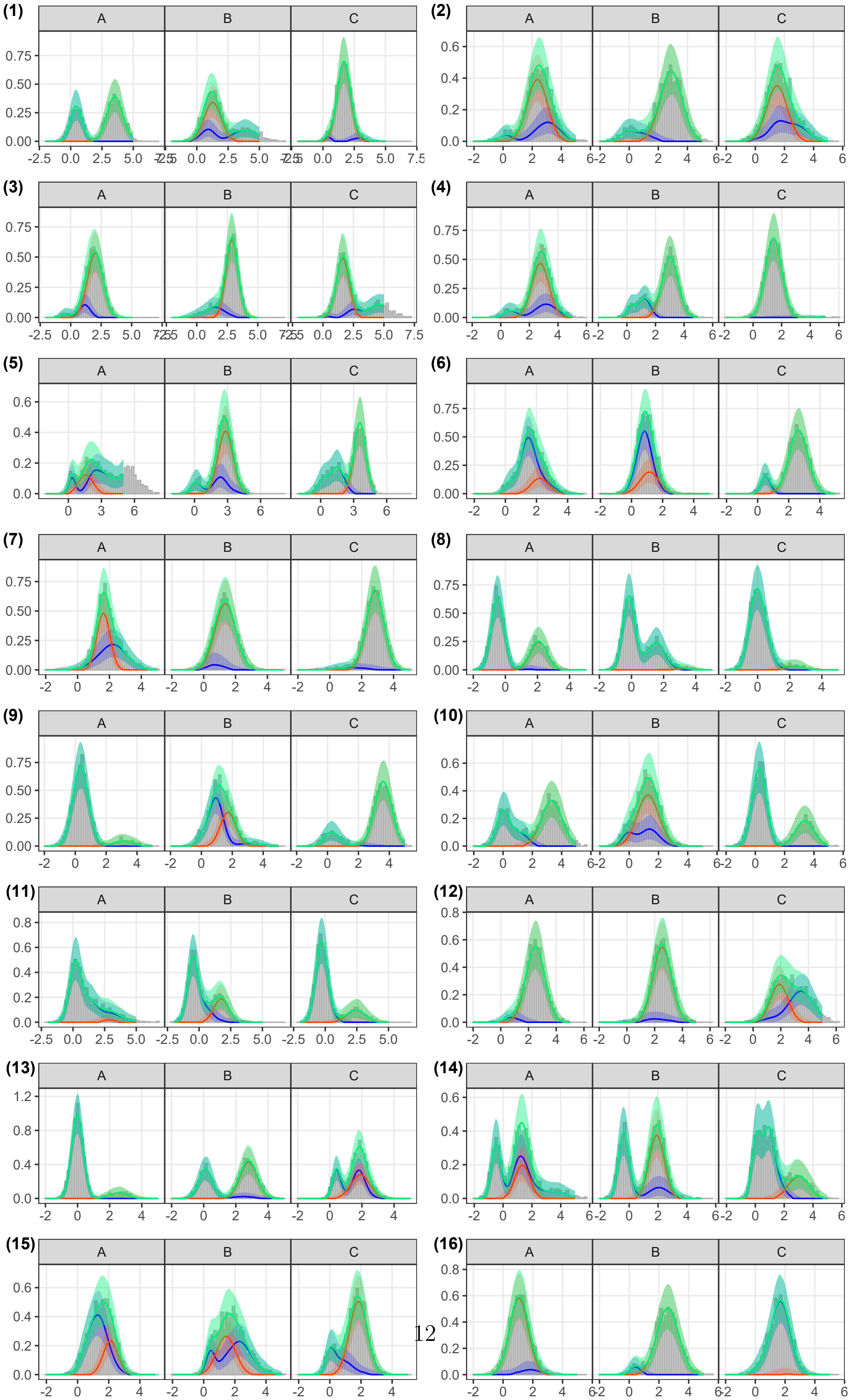


Figure S6 **Estimated vs true Gaussian parameters in 3D, 5D and 7D simulations.** Coloured points and lines show the median and 95%CrI model parameter estimates compared to the true values in 50 simulations. Columns indicate the number of pathogens (3, 5 or 7), while rows represent each parameter. The black dashed line indicates where estimated values would equal true values. Blue points and lines highlight the model estimates where the true value was contained within the 95%CrI and orange highlight those that did not contain the true value within estimated 95%CrI.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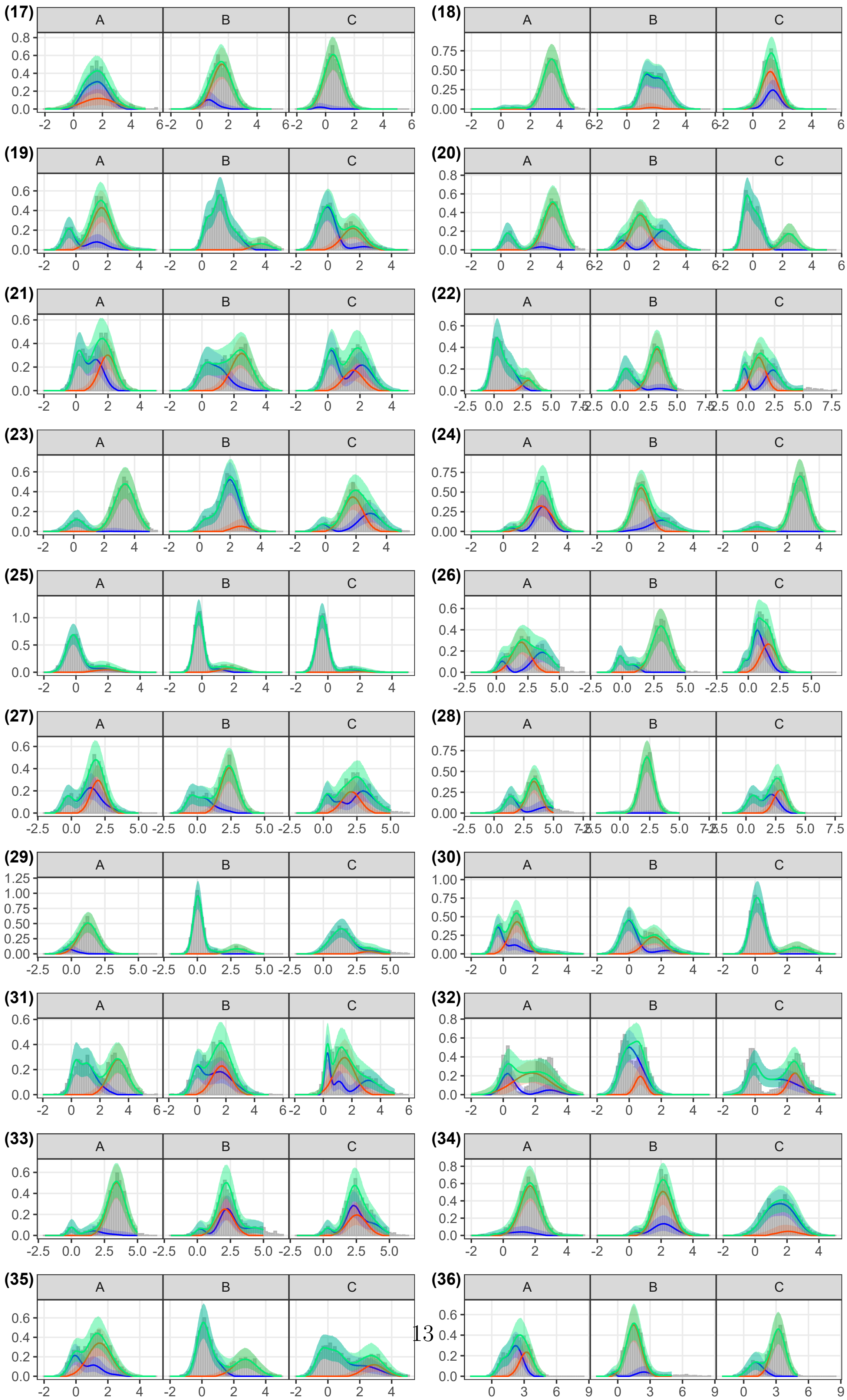
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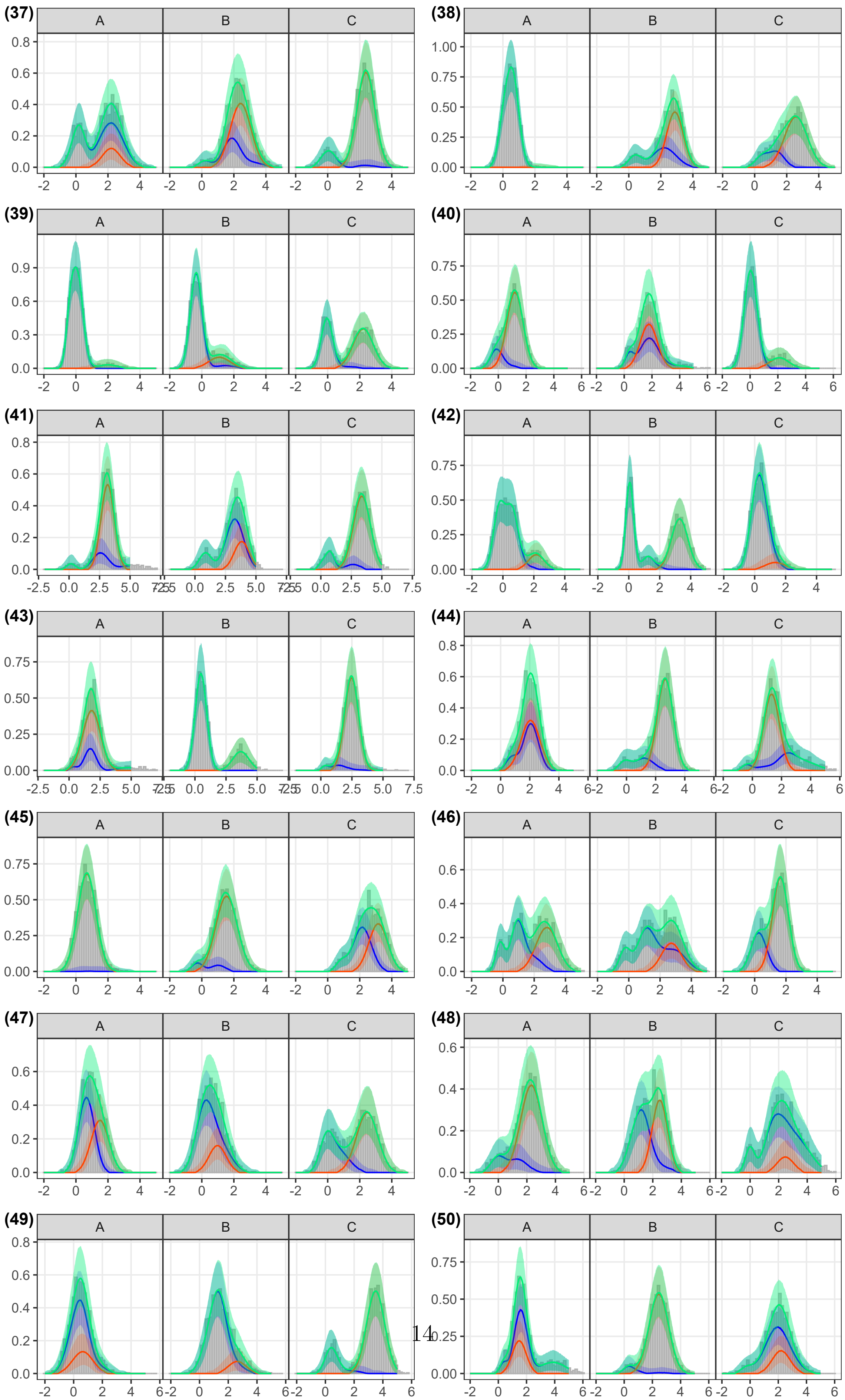
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Figure S7 **Model fits to 3D simulations.** Grey bars show the density distribution of simulated antibody titers for each of the 50 simulations, for pathogens A, B and C. The lines and shaded ribbons show the median and 95%CrI model reconstructed titer distributions. Blue and orange indicate the reconstructed distribution of negative and positive titers to each pathogen respectively, while green shows the overall reconstructed titer distribution (combined across all components).

2.3 Varying infection prevalence

In this section we fit two model versions to simulated data for 2 related pathogens where infection prevalence varies across 3 locations of equal sample size. We compared the performance of a base model that estimates only a single overall prevalence per pathogen and that of a location model that estimates pathogen-specific prevalence for each location. We found good consistency between estimated and true overall infection prevalence for both the base and location models with adjusted R^2 values of 0.91 and 0.99, respectively. True values of overall infection prevalence values were within the estimated 95%CrIs for 80% and 100% of the base and location model estimates respectively as shown in Figure S8. In addition, 100% of location-specific prevalence estimates by the location model accurately recovered the true values (Figure S8). For the cross-reactivity parameters, ϕ, 91% and 96% estimates were accurately estimated by the base and location models respectively, shown in Figure S9. 84% and 92% of ρ_{00} and 84% and 94% of ρ_{11} estimates were accurately estimated by the base and location models respectively (Figure S9).

For μ_0 parameters, true values were recovered for 92% and 95% of estimates across the 50 simulations, while for μ_1 parameters true values were recovered in 92% and 98% of estimates for the base and location models respectively. In addition, 86% and 92% of σ_0 true parameter values, and 92% and 96% of σ_1 true parameter values were recovered by the base and location models respectively, shown in Figure S10. The location model outperformed the base model in all 50 simulations, as measured by DIC, WAIC and ELPD LOO. The median difference in DIC (base - location model) was 734.70 (range: 90.30 to 1967.9) and 729.73 (range: 90.76 to 1965.44) for WAIC. For the ELPD LOO metric the median difference (base - location model) was -364.9 (range: -982.71 to -45.38).

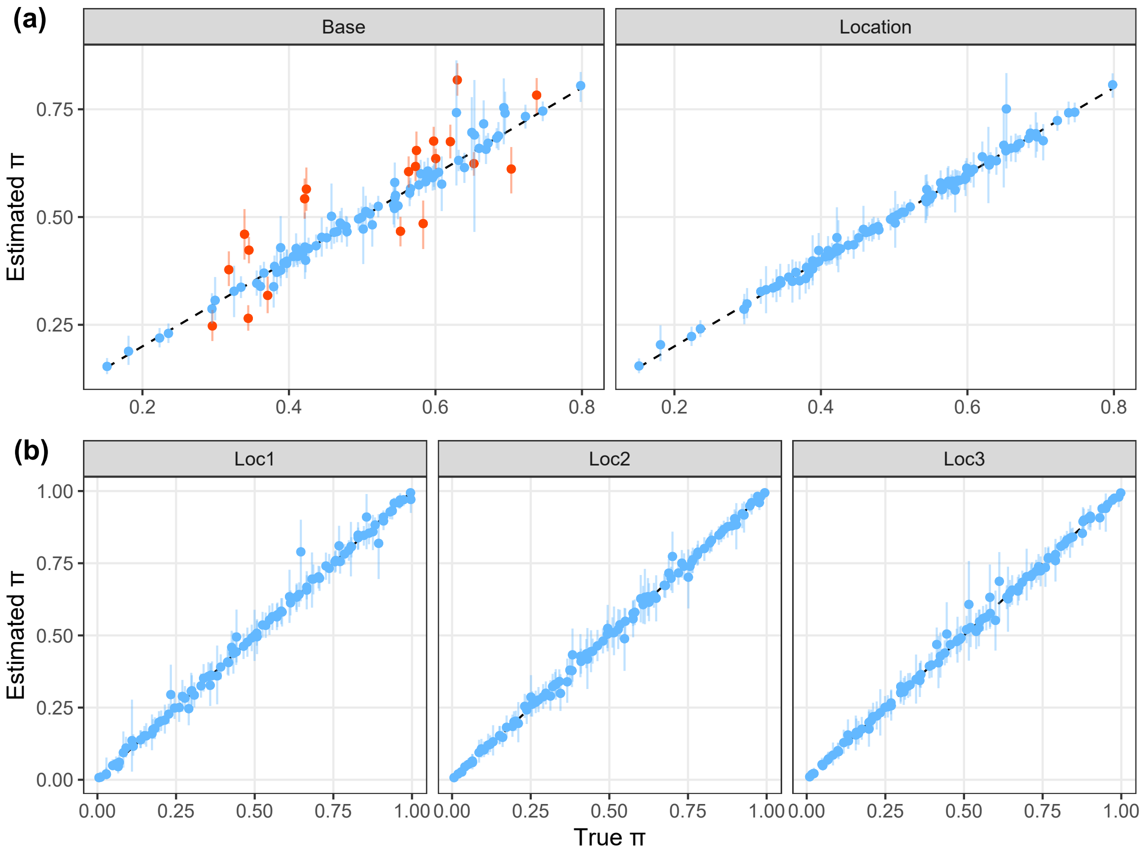

Figure S8 **Recovery of location-specific infection prevalence estimates.** Panel (a) shows the estimated vs true overall pathogen prevalence estimates from the base model and location model. Panel (b) shows the estimated vs true location-specific prevalence estimates from the location model. Coloured points and lines show the median and 95%CrI model estimates of the prevalence parameters, compared to the true values in 50 simulations. The black dashed line indicates where estimated values would equal the true values. Blue points and lines highlight the model estimates where the true value was contained within the 95%CrI and orange highlight those that did not contain the true value within estimated 95%CrI.

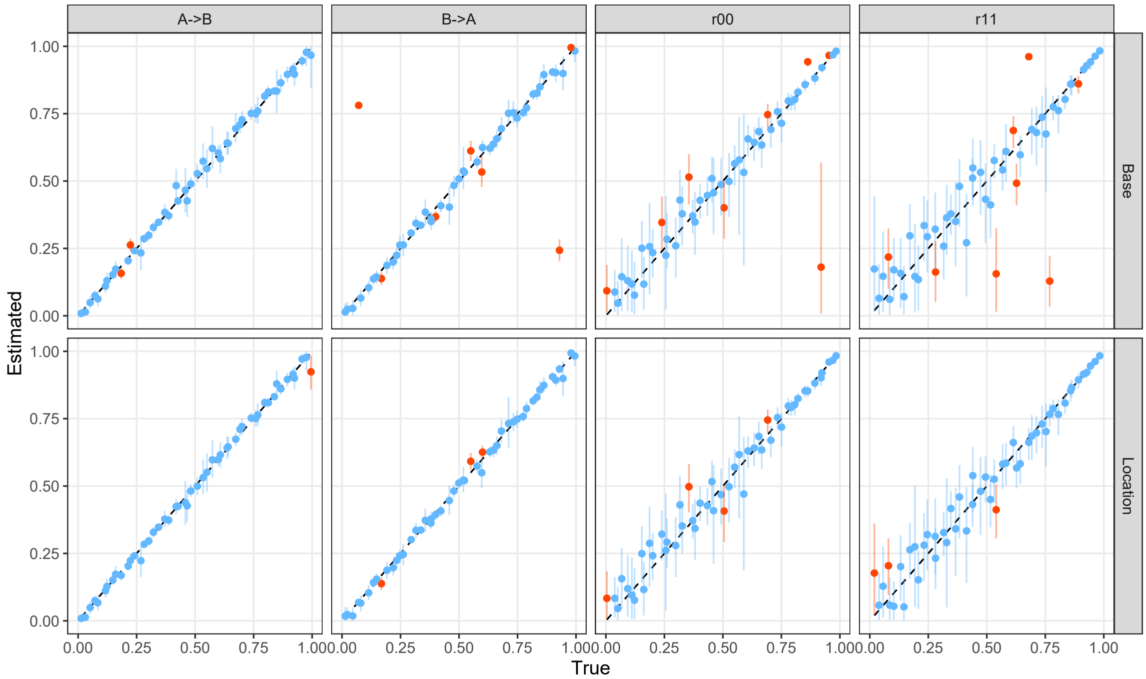

Figure S9 **Recovery of cross-reactivity and correlation parameters with heterogeneous infection prevalence.** Coloured points and lines show the median and 95%CrI model estimates of cross-reactivity and correlation parameters, compared to the true values in 50 simulations. The black dashed line indicates where estimated values would equal the true values. Blue points and lines highlight the model estimates where the true value was contained within the 95%CrI and orange highlight those that did not contain the true value within estimated 95%CrI. Panels A-¿B and B-¿A represent the estimates of relative titer increase from A to B and B to A, respectively. Panels r00 and r11 represent the correlation in the Gaussian distribution for infection statuses negative to both pathogens and positive to both pathogens, respectively. Row panels represent which model the estimates come from - the base model or location model.

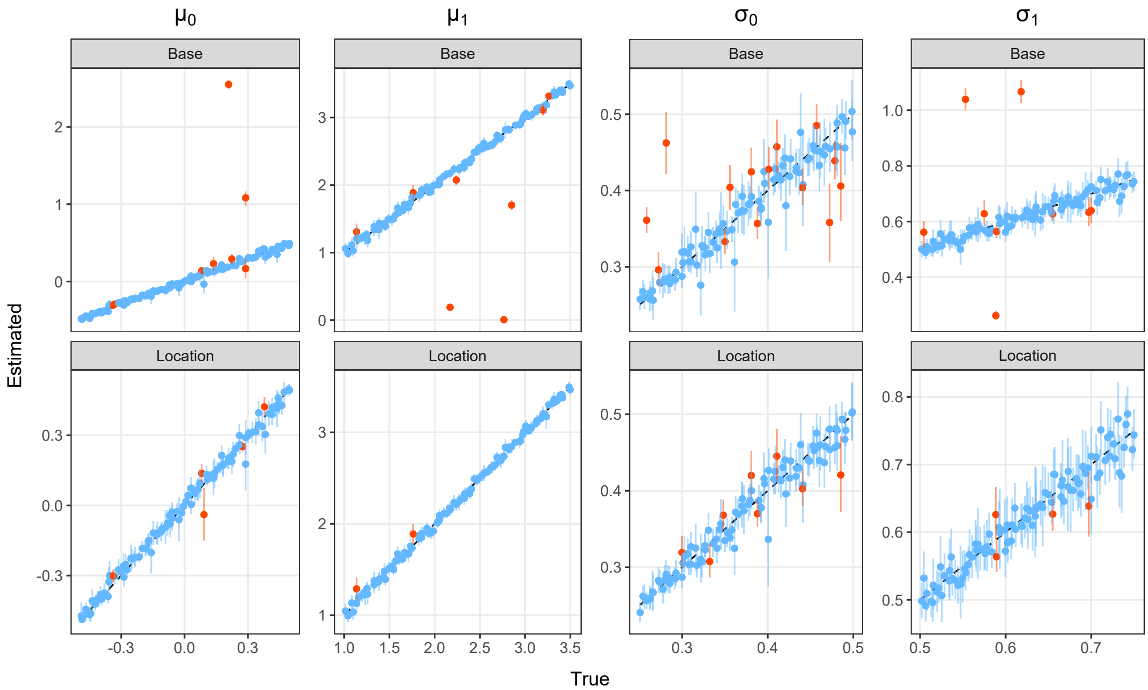

Figure S10 **Estimated vs true Gaussian mean and standard deviation parameters from location and base models.** Coloured points and lines show the median and 95%CrI model estimates of the μ and σ parameters, compared to the true values in 50 simulations. The black dashed line indicates where estimated values would equal the true values. Blue points and lines highlight the model estimates where the true value was contained within the 95%CrI and orange highlight those that did not contain the true value within estimated 95%CrI. Row panels represent which model the estimates come from - the base model or location model.

2.4 1D model biases

To understand the levels of bias that can occur when cross-reactive antibodies are present but not accounted for, we fit classic single-dimension (1D) mixture models independently to each of the two pathogens from the same two pathogen system simulations of the previous section. Low consistency between estimated and true infection prevalence values, π, was observed from these models with an adjusted R^2 of 0.40. True values of prevalence were contained within the 95%CrIs in 55% of estimates, shown in Figure S11, panel (a). We observed large overestimates of prevalence from the 1D models, coinciding with a general trend of underestimation of μ_1 parameters and overestimation of μ_0, shown in Figure S11, panel (b) and Figure S12. We found that the absolute bias in prevalence estimates is highest in simulations where both the level of prevalence of the unobserved pathogen is high and the level of cross-reactivity from infection with the unobserved pathogen against the observed pathogen is also high, shown in Figure S13. In contrast, high prevalence of the unobserved pathogen coupled with low cross-reactivity to the observed pathogen caused less bias in prevalence estimates. Similarly, high cross-reactivity from the unobserved to the observed pathogen coupled with low prevalence levels of the unobserved pathogen also resulted in reduced bias in prevalence estimates by the 1D model (Figure S13).

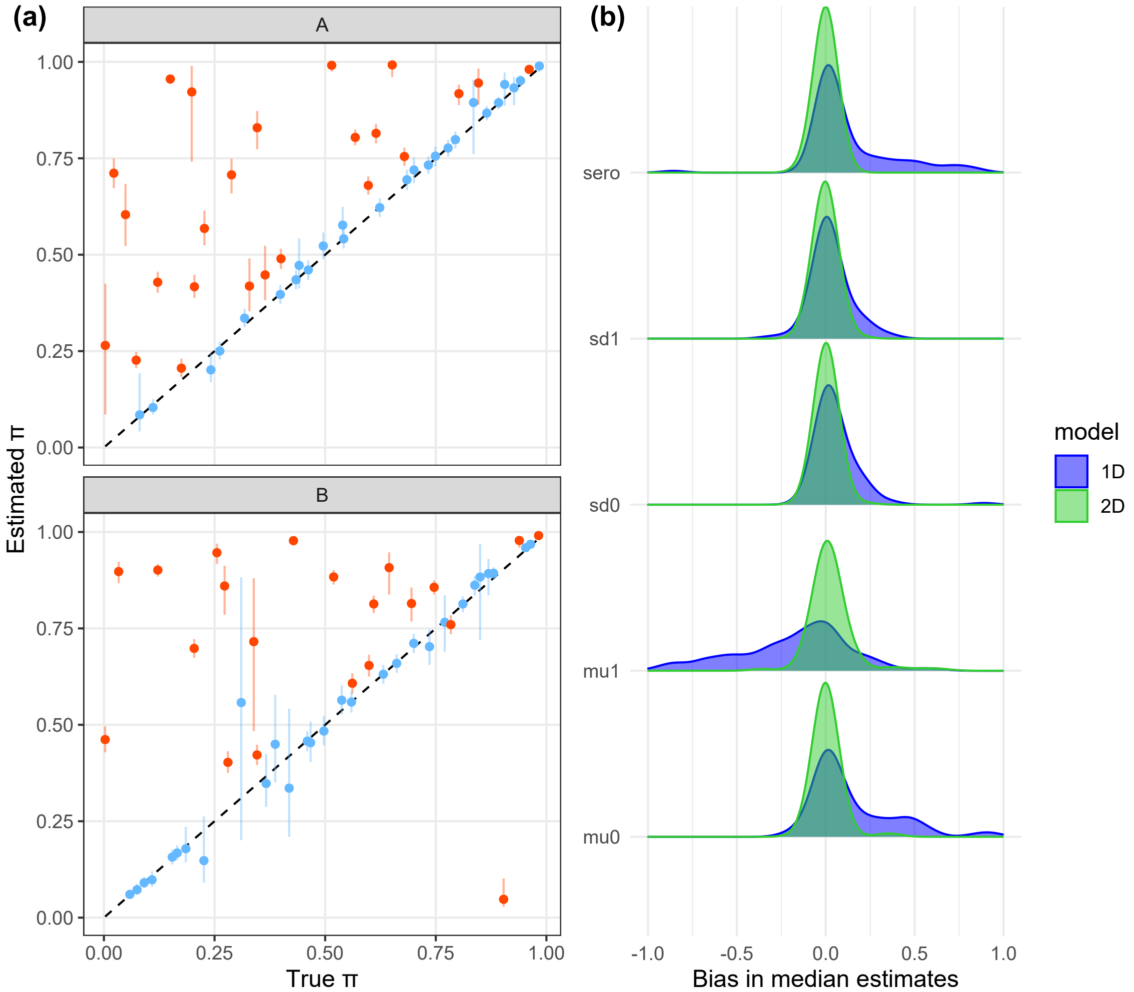

Figure S11 **Bias in 1D model parameter estimates.** Panel (a) shows the estimated vs true prevalence, π, from fitting classic 1D mixture models to 2D simulated data independently for pathogens A and B. Coloured points and lines show the median and 95%CrI model estimates of the π, compared to the true values in 50 simulations. The black dashed line indicates where estimated values would equal the true values. Blue points and lines highlight the model estimates where the true value was contained within the 95%CrI and orange highlight those that did not contain the true value within estimated 95%CrI. Panel (b) shows the density of bias (estimated - true) in model median estimates for all parameters for both the 1D (blue) and 2D models (green) for 50 simulations.

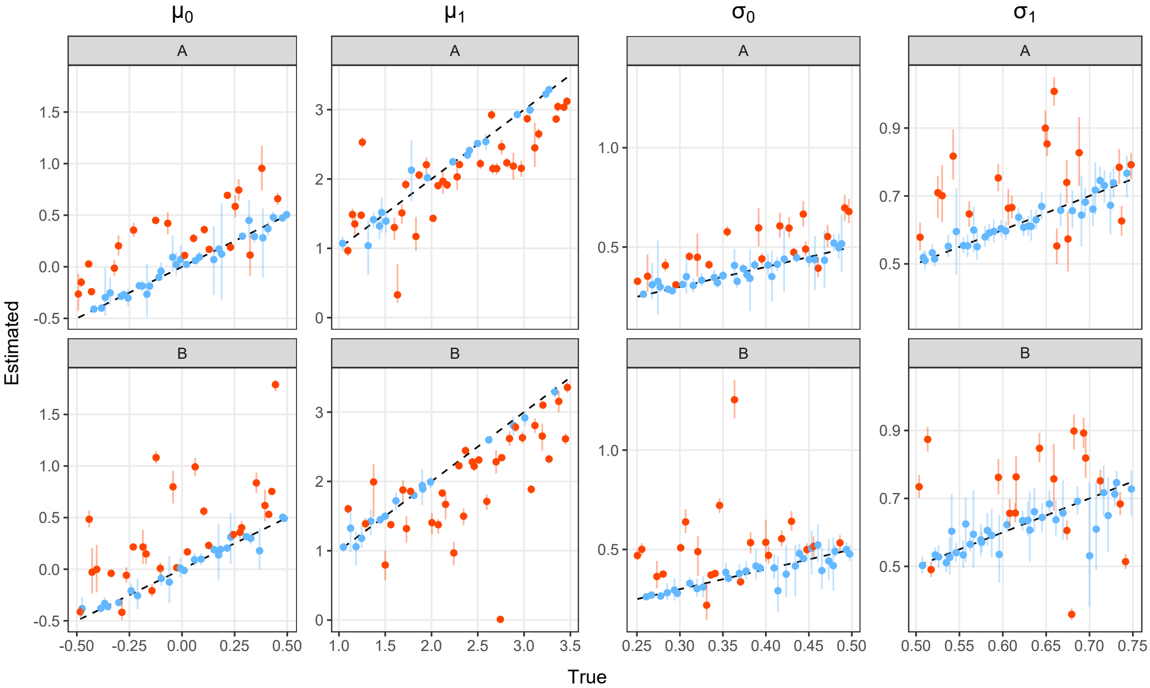

Figure S12 **Estimated vs true Gaussian mean and standard deviation parameters from 1D models.** Coloured points and lines show the median and 95%CrI model estimates of the μ and σ parameters, compared to the true values in 50 simulations. The black dashed line indicates where estimated values would equal the true values. Blue points and lines highlight the model estimates where the true value was contained within the 95%CrI and orange highlight those that did not contain the true value within estimated 95%CrI.

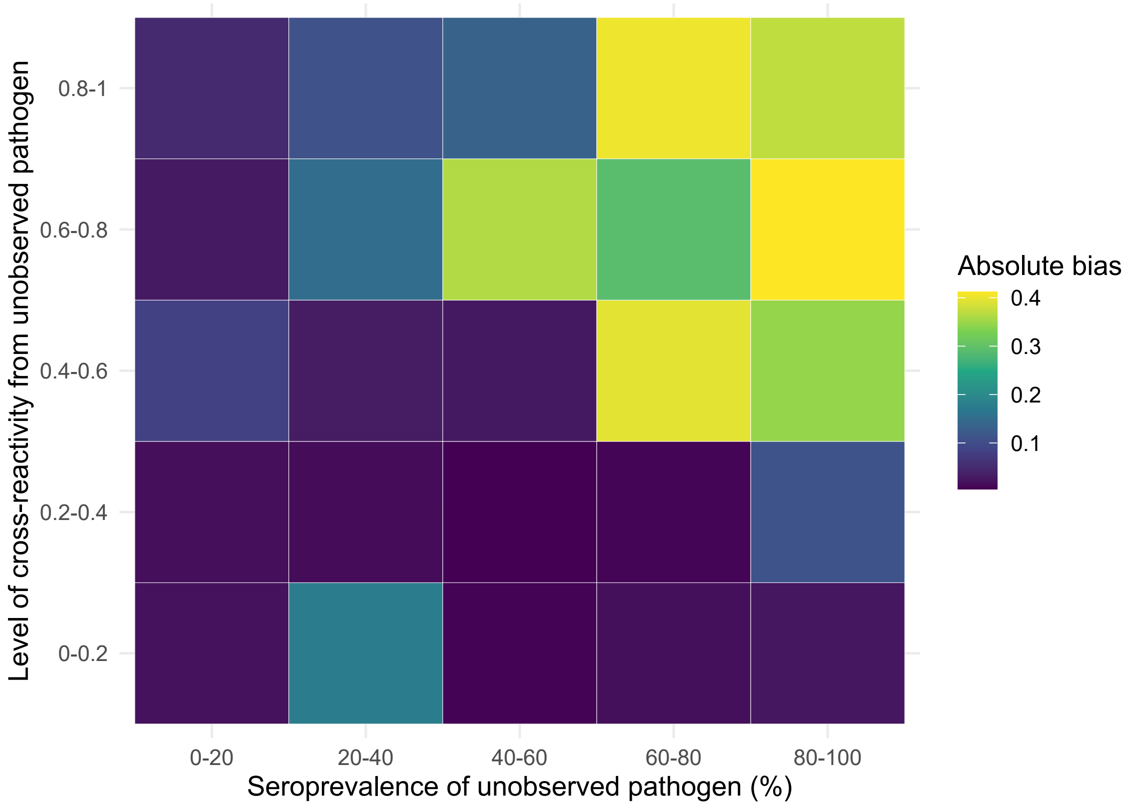

Figure S13 **Bias in infection prevalence estimates from 1D models.** Coloured tiles show the absolute bias in prevalence estimates by 1D models for varying levels of prevalence of the unobserved pathogen as well as varying cross-reactivity from infection with the unobserved pathogen against the observed pathogen.

2.5 Determining pathogen presence vs absence

In this section we explored the ability of the model to differentiate between present and absent pathogens. Using the same 2 and 3 pathogen simulations as in the previous sections, we set the prevalence of one pathogen to zero and re-simulate the data with N=1,500. We then fit the same base model as before to each dataset as well as the "absent model" which assumes the absence of the truly absent pathogen. Only one 3D simulation was excluded from summary results due to non-convergence of model chains. High consistency between median estimates and true prevalence of the present pathogens was observed for both models with adjusted R^2 values of 1.00 and 1.00 for the base and absent models respectively in both 2D and 3D simulations. Estimates of prevalence contained the true value within 95%CrIs for 100% of simulations for both the base and absent model in both 2D and 3D simulations, shown in Figure S14. For the absent pathogen, the base model estimates near zero infection prevalence for the majority of simulations, with a median estimate of 0.001 (range: <0.01-0.09) across simulations for the 2D system and a median estimate of 0.002 (range: <0.01-0.02) across the 3D simulations (Figure S14).

Cross reactivity parameters from the present pathogens to the absent pathogen were well recovered in both 2D and 3D simulations, shown in Figure S15. Cross reactivity estimates by the base model from the absent pathogens to the present pathogens, returned the prior in the majority of simulations (Figure S15). Correlation of negative titers, ρ_{00} was also well recovered in most simulations. Correlation of positive titers, ρ_{11} could only be well recovered in 3D simulations where a signal from the 2 present pathogens allowed it to be identified (Figure S15). In 2D simulations, as expected, the base model could not identify this parameter and returned the prior. High recovery of true Gaussian mean and standard deviation parameter values by the base and absent models were also observed for both 2D and 3D simulations. As expected, the base model returned the priors for μ_1 and σ_1 parameters that related to the absent pathogens.

To compare model performance between the base and absent models we calculated the difference in metrics between models (base model metric - absent model metric) as well as the LIPpp and LIPpc. The absent model was preferred to the base model by DIC in 45/50 2D simulations and 42/49 3D simulations, shown in Figure S16. Comparing model performance by WAIC, 44/50 simulations preferred the absent model in 2D simulations and 46/49 in 3D simulations. By ELPD LOO, 44/50 and 46/49 simulations preferred the absent model to the base. The median difference in log-likelihood values for 2D simulations was -0.74 (range: -1.19 - 5.23) and -0.79 (range: -1.58 - 5.21) in 3D simulations (Figure S16). Differences in DIC ranged from -2.33 to 8.01 with a median of 3.00 in 2D simulations, compared to a median of 3.43 in 3D simulations, ranging from -4.64 to 11.08. Median differences in WAIC were 1.72 (range: -6.93 - 3.05) and 1.85 (range: -6.43 - 4.93) in 2D and 3D simulations respectively. The median LIPpp for base compared to absent models across 2D simulations was 0.01 (range: -0.27 - 0.14) and 0.00 (range: -0.02 - 0.05) in 3D simulations, shown in Figure S16, panel B. The median LIPpc for base compared to absent models was -0.02 (range: -0.54 - 0.28) in 2D simulations and 0.00 (range: -0.02 - 0.06) in 3D simulations.

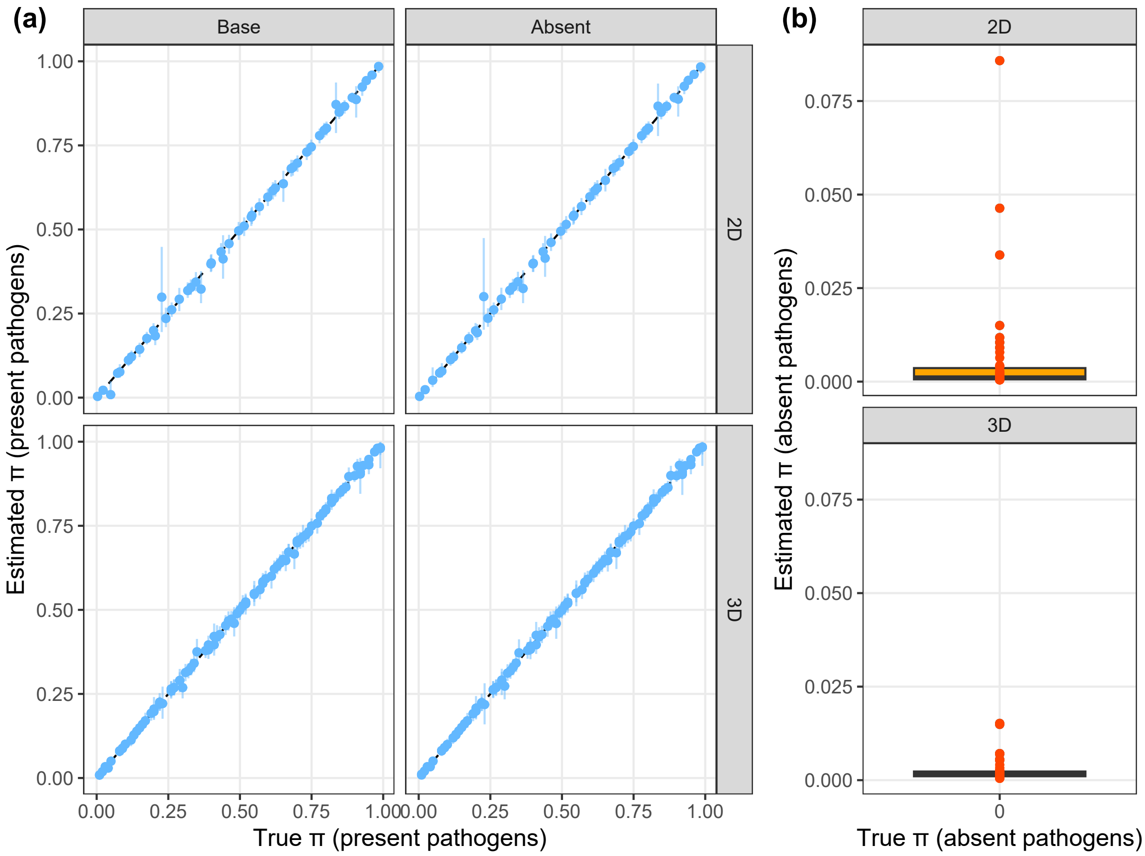

Figure S14 **Recovery of infection prevalence parameters with absent pathogen.** Panel (a) shows the estimated and true infection prevalence values π of present pathogens estimated by the base or absent model (column panels) for 2D and 3D simulations (row panels). Points and lines indicate median and 95%CrI estimates. Blue points and lines highlight the model estimates where the true value was contained within the 95%CrI and orange highlight those that did not contain the true value within estimated 95%CrI. The black dashed line indicates where estimated values would equal the true values. Panel (b) shows a box plot of median prevalence estimates by the base model for the absent pathogen (where true prevalence equals zero). Orange points represent the median estimates from individual simulations.

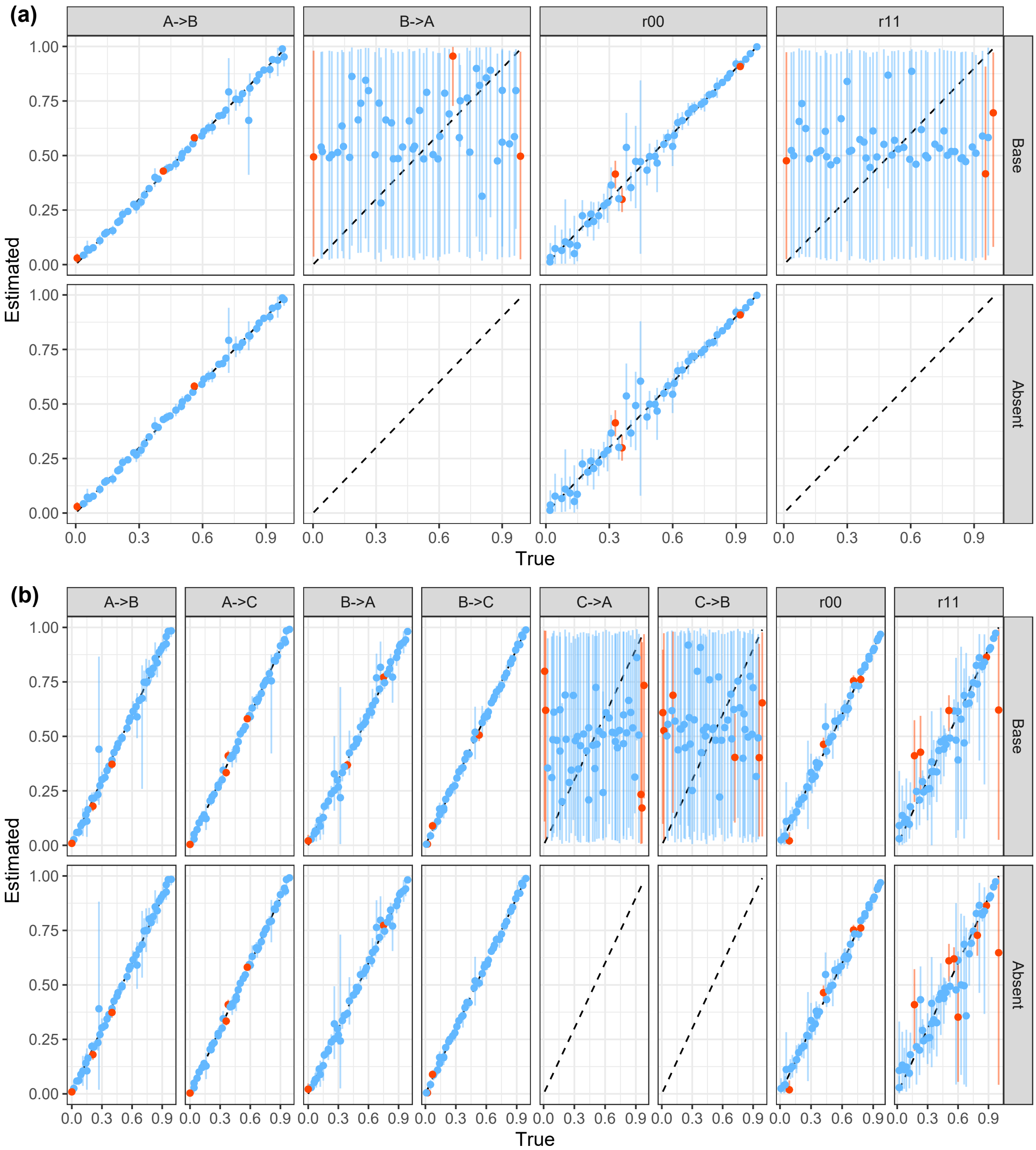

Figure S15 **Recovery of cross reactivity and correlation parameters with absent pathogen.** Panel (a) shows the estimated vs true values of cross reactivity parameters, ϕ, and correlation parameters, ρ, in 2D simulations. Panel (b) shows the same but for 3D simulations. Points and lines indicate median and 95%CrI estimates. Blue points and lines highlight the model estimates where the true value was contained within the 95%CrI and orange highlight those that did not contain the true value within estimated 95%CrI. The black dashed line indicates where estimated values would equal the true values. Column panels indicate individual parameters and row panels indicate the model used for fitting.

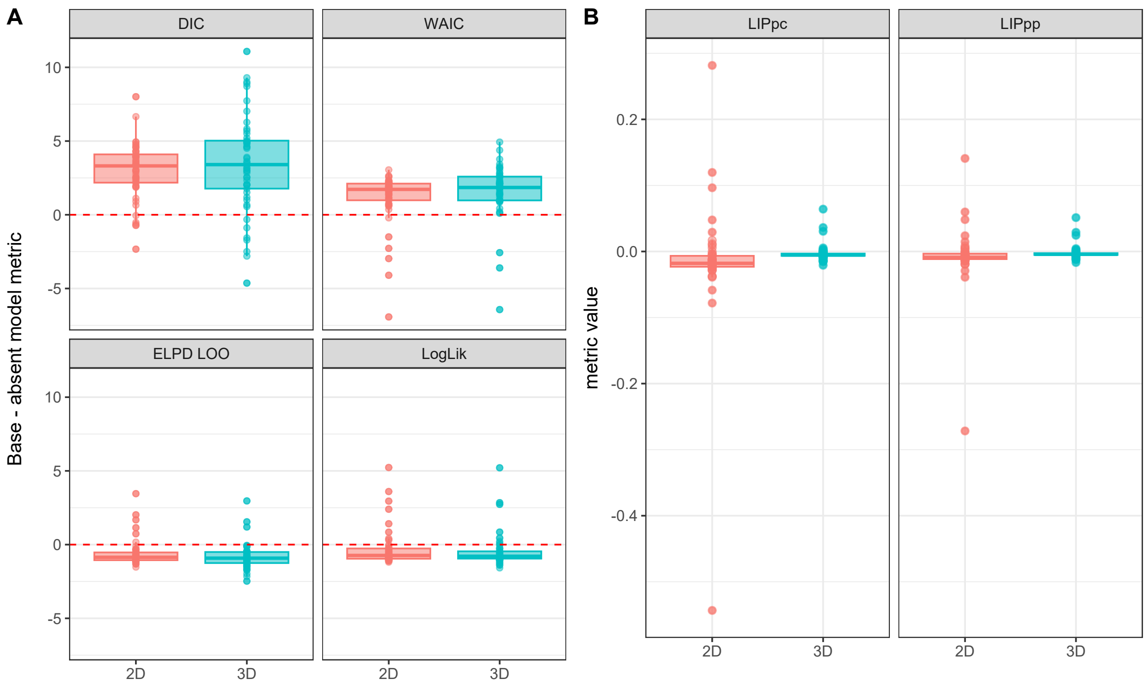

Figure S16 **Absent vs Base model performance metrics.** Panel A shows box plot summaries of the difference in model metrics (base minus absent model) for 2D and 3D simulations. Coloured boxes show the median and interquartile range of values while points show the individual values. Red dashed lines highlight a difference of zero, where base and absent model performances are equal. Panel B shows the likelihood increment percentage per parameter (LIPpp) and likelihood increment percentage per component (LIPpc) in 2D and 3D simulations. This calculates the percent increase in log likelihood per parameter or component for the base model compared to the simpler absent model.